Integrating Blockchain Technology in 5G enabled IoT: A Review

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***Abstract*—In the digitized world, Internet of thing(IoT) is growing rapidly and most of the devices are connected by internet technology. IoT connected more than 5 million devices across the world and it needs high power internet access, data transfer security, decentralization, low latency etc. When number of interconnected devices increases, IoT technology needs high-band with data communication and 5G can resolve network related issues in the IoT devices. However 5G enabled IoT devices suffer lot of security related problems due to centralization. This paper review the security, performance, network issues in the IoT devices based on 5G technology and integration of Blockchain technology in 5g-IoT devices.**

***Index Terms*—5G, IoT, Blockchain, SDN, Smart City, IoV**

I. Introduction

5G, IoT and blockchain [1] are three different independent techniques but together they can shape our future. 5G is the latest cellular network technology and expected to be implemented 2020, but some countries are already tested this new technology. The data transmission rate of 5G is 10Gb per second and it offers faster internet, lower latency and greater interconnection. These advantages of 5G is enable to interconnect more IoT devices together. The main applications of IoT are in healthcare, smart cities, smart cars, smart farming, smart homes, smart TVs, furniture items and vacuum cleaners etc. Smart cities and smart homes can save energy up-to 15 to 20% and as parallel to the advantages, it has lot of security issues in the smart devices. Integration of blockchain technology with IoT can solve the security issues in the smart devices. In Blockchain technology, data blocks are linked together with previous block address and it is more secure than traditional client server system. The main features of blockchain technology are decentralization, peer-to-peer network, enhanced security, smart contract, consensus and distributed ledger. The security issues in the IoT devices can solve to some extend by the integration of blockchain technology with 5G and IoT.

This paper reviews the advantages and security issues of the integration of three technologies such as Blockchain, IoT and 5G technologies. A categorical study of recent research works in the field of fifth generation of wireless technology is carried out in this paper. This paper is organised as in section II about 5G technology, section III about 5G and IoT, section IV about technical and security issues in 5G IoT, section V about blockchain technology, section VI about Blockchain enabled 5G-IoT, section VII about open challenges and security issues in blockchain enabled 5G-IoT and section VIII is conclusion.

II. 5G Technology

5G technology [2] is the future of next generation which use CDMA, BDMA and millimetre wireless technologies with speed of higher than 100MP. The important technologies behind the implementation of 5G are D2D communication, Massive Multiple Input and Output(MIMO), smart devices, millimeter wave and GFDM [3].

A. Evolution of mobile network

First analog mobile communication was started in US at 1950's. After 3 decades first generation (1G) mobile was launched. After the invention of microprocessor, second generation of mobile communication (2G) technology was launched with digitized properties by using 1's with SMS capability. After few years, GPRS facility was added to the 2G network and able to browse internet in the mobile. Third generation (3G) came in the 21st century with the capable of voice and messaging with satisfactory speed. Fourth generation(4G) was launched few years back with super fast rate of internet communication in 4G enabled devices. Now a days internet usage is not only in mobile devices but also popular in electronic devices such as fridge, television, cars etc. In-order to achieve uninterrupted network connection and to connect more devices, it needs high speed internet and bandwidth. This is the initial steps for the Fifth generation (5G) technology and the connectivity speed is up to 25Mps. The main advantages of 5G are high data transmission, lower latency, support virtual private network , high bandwidth etc. The main application area of 5G are in health sector, mobile industry, IoT, artificial intelligence etc.

5G will connect each and every corner of human life with technology through communication networks. So it is important and need to be protected sensitive data from cyber attack. The main security threats in 5G technology's are DoS attack, saturation attacks, scanning attacks, timing attacks, man in the middle attack, user identity theft, hijacking attacks

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etc. So when 5G technology is integrating with Iot and blockchain application , it needs a security architecture [3] .The following sections are discussing about IOT, blockchain and its integration.

III. 5G AND IOT

Internet of things [4](IoT) is the interconnection between physical devices, vehicles, building etc and data sharing by electronic or sensors embedded in these devices. As parallel to the increment of IoT devices, its need high speed internet connection and secure environment. With the help of 5G technology IoT can connect billions of devices without the human interaction between them. The main requirements of IoT are high data rate, virtual reality or augmented reality, high definition video streaming and network scalability. Some IoT application needs lower latency, reliability, high coverage, highly protected security, low energy solution and massive number of device connectivity.

A. 5G IoT Architecture

5G IoT architecture [5] consists of business layer, application layer, support layer, connectivity/edge computing layer and recognition layer. IoT physical object such as smart phones, smart wearable, smart vehicles etc and hardware devices such as sensors, controllers or other data capturing devices from IoT are includes in the recognition layer. The main functionality of this layer is collect information from IoT devices and recognise IoT environment. Communication protocols and networks are defined in the connectivity/edge computing layer and main functionality of this layer is connectivity. Support layer supports the above and below layers for connectivity and it includes cloud computing powers, data analysis ability, and other intelligent computing ability. Application layer supports different applications such as smart home, smart grid, smart healthcare, smart transportation, smart wearable, etc. Business layer mange all IoT systems. In each layer there are lot of security issues like hardware vulnerabilities, network security and software security. The main service requirements for 5G IoT devices are network function virtualisation (NFV), scalability, mobility control and smart services. In 5G-IoT application, NFV provides scalability and flexible in network function.

B. Software Defined Network

5G is the largest heterogeneous network [6] and the features are high data rate, mobility, less transmission delay, energy consumption, high data coverage etc. For this advance features, 5G technology needs an efficient network management system and agility, flexibility, network remote controlling and programmable networks. SDN is the better choice for it. Control and management are the main function of SDN in 5g technology. It is implemented in the 5G network through Network Function Virtualization(NFV) [6]. The architecture of SDN is shown in Fig 1.

In the SDN architecture, there is clear separation between data plane, control plane and management plane. For managing complex network like 5G this SDN architecture is essential. SDN also allow network function Virtualization(NFV). Network engineers considered open flow is standard for SDN because softwarization is important in networks.

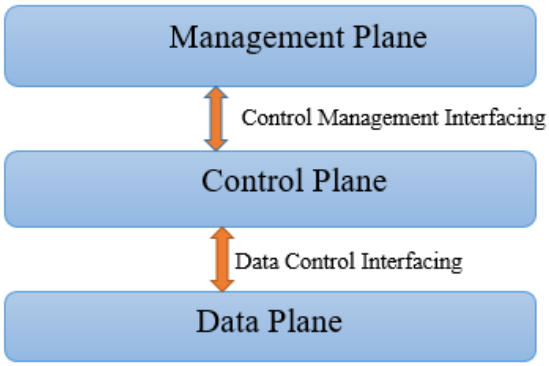

Fig. 1. SDN Architecture

1) Security Challenges in SDN based 5G technology: SDN architecture [6] [7] can improve network efficiency in 5G technology as parallel it has lot of security problems in communication channel and is vulnerable due to lot of reasons. The main security challenges in SDN based 5G technology [7] are Spoofing attacks, man in the middle attack, eavesdropping, malicious attacks and unauthorized access and it is due to weak authentication between controller and application.

C. Network Function Virtualization(NFV)

5G technology allows physical network is split into different virtual network and it also support different radio access networks. NFV [7] helps virtualization and network slicing in 5G network. The main advantages of the network slicing of NFV in the 5G technology are mobility, lower latency, resiliency, security and availability.But the trust maintenance in virtualized NFV is challenging. The main security challenges of NFV in 5G technology are confidentiality, integrity, authenticity and non repudiation. Security problems and configuration error are the main issues in dynamic nature of the NFV. Moreover the security attacks in NFV are spoofing, sniffing, Dos attacks, flood attacks , side channel attacks etc.

IV. TECHNICAL AND SECURITY ISSUES IN 5G-IOT

IoT is the interconnection between computing devices, digital machines, mechanical devices, or any other object that can transfer data across through network without human interaction or Computer. According to Gartner forecasts, in 2020 the world will use 20.4 billion devices. When the devices connectivity increases it faces lot of technical issues and cyber attacks [8]. The main technical and security issues [7] in 5G IoT architecture are scalability, network management, interoperability, lack of standards, latency, heterogeneity, security assurance, identity, privacy concerns and authentication. Some

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sort of security related issues can solve by integrating block chain technology with IoT application [5].

V. Blockchain Technology Overview

Blockchain [9] [10] [11] is the growing list of blocks or chains linked together by cryptographic hash function. It is fundamentally distributed database of records. It has a public ledger for handling all transaction related to different parties and it is decentralized. Consensus mechanism is the backbone in the blockchain technology and each transaction is verified by using consensus algorithm and after verification it is recorded in the ledger. The transaction details recorded in the ledger is permanent and cannot erase or change recorded transaction details. The main idea of blockchain technology [6] is started from the paper title Bitcoin: A Peer-to-Peer Electronic Cash System written by anonymous person/group of person named as Satoshi Nakamoto at 2008. The most popular application of Block chain technology is the bitcoin-decentralized peer to peer digital currency.

A. Working of Blockchain

Blockchain is he group of blocks linked together with cryptographic hash function of the next node. When a new transaction is requested to the blockchain, first they a create a node for representing the transaction. Then this node or block is broadcast to all other nodes and all nodes validate the newly created block. If all nodes validation are completed , the new node is added in the blockchain ledger and transaction of the block is recorded.

B. Types of Blockchain

Blockchain technologies are mainly classified [6] as Permissioned(Private), Permissionless(Public) and consortium. Bitcoin, Ethereum are examples of public block chain and it is transparent to all. Private blockchain is mainly for specific organisation and only authorized persons can validate,issue and read transaction details. Transparency, decentralization and anonymity are the features of private block chain. Blockstack and Multi-chain are examples of private blockchain. Consortium blockchain[1] is semi-decentralized, controlled by group of authorized persons with permission other than single entity and consensus process is handle a group of pre-approved nodes in the network. Hyper-ledger, Corda and Quorum are example of this network.

C. Characteristics and Application

The two main characteristics [6] [12] of blockchain technology are distributed, decentralized, smart contract, trust, transparency, immutability, security and anonymity [13]. Due to these features most of the online transactions are come up with blockchain technology. The main advantage is there is no need for intermediate in these transaction and all the history of transactions are recorded. Smart contract is set of computer program stored in blockchain and it is executed automatically when some condition is met.The Blockchain technology has application in both financial and non financial areas. Blockchain application in non financial are notary public, music industry, signed document validation, decentralized storage, decentralized IoT and blockchain based anti counterfeit solutions.

VI. Block chain enabled 5G IoT

Now a days digitization is increase and as parallel usage of IoT devices also increases. According to the Gartner [14] forecasts it becomes 20.4 billion connected devices at the 2020 years. The main challenges of IoT are decentralization, compatibility , privacy and vulnerability in security problems. Most of these challenges can solve by integrating blockchain technology with 5G enabled Iot. Blockchain is distributed ledger and main highlight of this technology is decentralization, it removes the intermediate in between the transaction. The integration of blockchain in IoT helps to remove centralisation and become the transactions are more secure ,autonomous and transparent. In this architecture general ledger is blockchain and it keeps all messages between smart devices as trusted. The main advantages of decentralised IoT [12] [14] are

- **Enhanced Security**:
  Decentralized network free from unauthorized attacks, vulnerability and cloud based data attacks .In Blockchain technology cyrptogtraphic hash function is use for securing communication between nodes, so it is difficult to hack the authorization details.
- **Free from Tampering and Fraud activity** :
  Asymmetric cryptography is used in distributed ledger to timestamp and store data are immutable. So this decentralised Iot is free from tampering and other fraud activities.
- **Affordable and trust**:
  By using decentralized blockchain technology and storing data in distributed ledger helps to remove vulnerability in IoT devices and it becomes more affordable. Blockchain technology uses smart contract and consensus algorithm for data transaction and it becomes more trust between the two parties.
- **Autonomous**:
  Using Blockchain technology in IoT , all devices become autonomous and it fully remove the central authority and all financial transactions can process without the central authority.

A. Blockchain Application in 5G enabled IoT

In the digitized world blockchain application become more powerful due to the decentralized behaviour. In the figure 2 shows Blockchain Application in 5G enabled IoT.

- **Smart City**:
  Smart city [15] [16] uses centralized services and interconnected networks it have lot of advantages and as parallel the potential threats are security and privacy. For securing the smart city application integrating blockchain technology with Iot is best solution. A security framework of blockchain enabled Iot is

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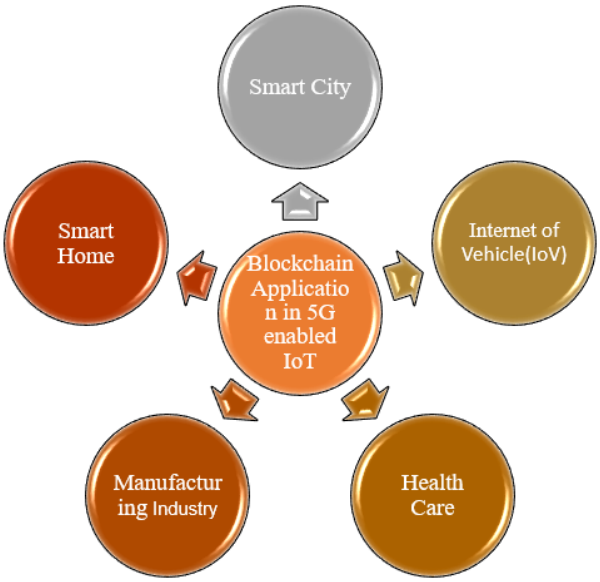

Fig. 2. Blockchain Applications

proposed kamanashis [14] and this framework solve the issues of reliability, scalability, security, centralization and efficiency.

- **Smart Home**:
  Smart home [15] [16] consists of smart door lock, smart lighting, video surveillance , smart thermostat and smart parking. The smart home needs the infrastructure as network connectivity, sensor devices enabled by IoT and mobile application for remote connection. In the smart home system, smart lock is very important because it protected the home from unauthorized user entry. If the smart home system uses central server there have lot of security problems such as unauthorized , forging ect. For avoiding these security problem Han et al [17]Ali Dorri [18] proposed blockchain enabled smart lock system and it gives the solution for non-repudiation and integrity. 5G enabled IoT also provides fast intrusion detection and low latency.
- **Manufacturing Industry**:
  Different manufacturing companies handles big-data and the main issues in the centralised manufacturing companies are data fragmentation, interoperability, upgrading problems of IoT firmware and security issues. Manufacturing industry becomes smart industry by the application of Blockchain enabled IoT and decentralization provides better security and privacy protection.
- **Health care**
  In the internet world healthcare [7] [15] application has more advanced and doctor can monitor patients health condition through remotely by the help of smart devices enabled by IoT. The main disadvantages of centralised IoT devices are unauthorized access and an attacker can easily attack the medical devices through vulnerability in the devices and it is very dangerous to the life of patients. To solve these security threats different authors proposed blockchain enabled IoT in healthcare applications. For more secure data transaction between medical devices and health data accessing Sun et al. proposed attribute based signature scheme and it is based on decentralized blockchain technology.
- **Internet of Vehicles(IoV)**
  IoV [7] [16] is the network between vehicles and internet. Main three types of IoV are Vehicles to Vehicle, Vehicles to infrastructure and Vehicles to cloud. Car connected to car, car connected to bus etc are includes in the application of vehicle to vehicle. In vehicle to cloud application is centralized and information about accident, vehicles private information all are stored in the cloud database. The main security challenges in IoV are sharing data among vehicles and storing information in cloud data-storage. Block chain enabled 5G IoT can solve most of the security related problems in message passing between vehicles or infrastructure by Proof of Work or smart contract and it also utilized energy efficiently. The main highlights of integrating these three technology together are enhanced security, integrity, authenticity and decentralization.

VII. Open Challenges and Security Issues in Blockchain enabled 5G IoT

As parallel to the advantages blockchain technology, it has some limitation when integrating this technology to 5G enabled IoT. Some of the main challenges [18] are energy consumption, scalability, slower data transaction, lack of standardization, low storage capacity and low computational power. As per the different statistics the number of IoT devices increases and it needs more battery power and energy for processing. In blockchain technology block mining is very computational expensive and energy needed. In blockchain technology data's are stored in the IoT nodes and it needs more storage capacity and high computational power. This is one of the challenges of IoT devices. Some security issues in blockchain technology are Dos attacks and trust between the two nodes. Data in the Blockchain technology are anonymous but in the case of IoV needs transferring personal information and other data .It is also somewhat problem in the blockchain enabled IoT applications. Due to the lack of standardization the devices communication through different ledgers are much more difficult.

VIII. Conclusion

In the digitized world the integration of different technology within one platform is very important and it can solve different technical, network and security related problems. IoT has different application in our daily lives and it needs high

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bandwidth and interconnection due to the increase of connectivity between devices. This paper reviewed the architecture, features and security related problems of three challenging technologies such as IoT, 5G and blockchain. As parallel to the advantages of blockchain enabled 5G IoT, the opening challenges of blockchain technology are scalability , interoperability, standardization, energy utilization and regulation. Some of these current issues can solve by integrating artificial intelligent with blockchain technology and still it is in open research problem.

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